# AHMED HARRAZ

**Backend Developer | Node.js, Express.js, NestJS, Fastify, TypeScript, MongoDB**

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## SUMMARY

Backend Developer specializing in Node.js, NestJS, and TypeScript with expertise in building secure, scalable RESTful APIs and production-grade authentication systems. Proven track record of optimizing performance (60% response time reduction), implementing enterprise-grade JWT authentication with refresh tokens and role-based access control, and refactoring complex codebases for stability and maintainability. Proficient in MongoDB, PostgreSQL, Redis, Docker, and AWS deployment.

## SKILLS

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| **Backend** | Node.js, Express.js, NestJS, Fastify, TypeScript, JavaScript, Socket.io |
| **Databases** | MongoDB, Mongoose, PostgreSQL, Prisma ORM, MySQL, Redis |
| **Auth & Security** | JWT, OAuth2, Bcrypt, Helmet, CORS, Rate Limiting, Refresh Tokens, RBAC |
| **Cloud & DevOps** | AWS (S3), Docker, Railway, Vercel, Git, Postman, Cloudinary, Node-cron |

## EXPERIENCE

**Backend Developer** July 2026 – Present
*Penta-K* *Onsite*

- Building and maintaining backend services with NestJS and TypeScript, including **call-99-backend** and a reusable **SaaS template** project.
- Designed and implemented two-network Docker Compose architecture for local development environments, resolving container networking and PostgreSQL/Redis connectivity issues.
- Studied and applied internal engineering standards including i18n, audit logging, and feature flags across the SaaS template.
- Collaborated on architectural improvements, including refactoring authentication flows and removing legacy 2FA implementation.

**Backend Developer** November 2025 – June 2026
*Travolic* *Hybrid*

- Architected and deployed **cheapest flights per month** endpoint handling hundreds of daily requests, enabling core travel search functionality for users comparing monthly flight prices.
- Implemented JWT authentication system from scratch for Travolic AI with refresh token rotation and role-based access control (RBAC), ensuring secure API access across multiple user tiers.
- Refactored large controller modules into modular, maintainable architecture, reducing code complexity and improving API stability while fixing critical bugs in flight search logic.
- Collaborated with cross-functional teams to design scalable backend architecture and optimize MongoDB queries for high-traffic endpoints.
- Integrated third-party travel data providers and payment gateways to enhance platform functionality.

## PROJECTS

**Call99 – On-Demand Automotive Services Marketplace Backend.** Developed a production-oriented automotive services marketplace backend using NestJS, TypeScript, PostgreSQL, and Redis, supporting car wash, roadside assistance, towing, car rental, provider onboarding, customer support, and administrative workflows. Implemented Paymob payment flows with HMAC-verified webhooks, idempotent processing, reconciliation, refunds, voids, provider subscription payments, and rental security-deposit handling. Built BullMQ workers for payments, notifications, email, media processing, cleanup, and subscription lifecycle jobs with retries and dead-letter handling. Delivered real-time

order tracking and customer/provider/support/staff chat using Socket.IO and Redis Pub/Sub, with session revalidation and recovery mechanisms. Implemented JWT authentication, refresh-token rotation, OTP flows, RBAC, audit logging, encrypted sensitive data, private S3 media, signed URLs, and asynchronous video-thumbnail generation.
*Tech: NestJS, TypeScript, PostgreSQL, TypeORM, Redis, BullMQ, Socket.IO, Paymob, Firebase FCM, AWS S3, Docker, Jenkins*

**Ganzo-Platform – Online Learning & Education Management Backend.** Developed and maintained a modular education platform backend serving students, parents, teachers, assistants, and administrators across courses, lessons, exams, subscriptions, payments, media, chat, and notifications. Built secure session-backed JWT authentication with refresh-token rotation, Google OAuth, RBAC, active-session enforcement, student login quotas, and detailed login-attempt history. Implemented an asynchronous video-processing pipeline using Redis, BullMQ, FFmpeg, and dedicated workers with retries, processing states, and cleanup workflows. Designed secure private-media delivery across AWS S3 and Cloudflare R2 using multipart uploads, expiring signed URLs, authorization checks, byte-range streaming, and per-user PDF watermarking. Developed authenticated Socket.IO chat and notification flows with Firebase push delivery, subscription lifecycle management, manual payment review, teacher-plan workflows, and transactional course-access activation.
*Tech: Node.js, TypeScript, Express.js, PostgreSQL, Sequelize, Redis, BullMQ, Socket.IO, AWS S3, Cloudflare R2, Firebase, Docker, Jenkins*

**Szissors – On-Demand Barber Booking API · NestJS Backend Platform.** Architected and built a production-grade freelance barber booking platform backend using NestJS, TypeScript, and PostgreSQL with a clean modular architecture supporting multi-role auth, barber onboarding, and real-time booking flows. Implemented repository pattern, custom guards, interceptors, and decorators for separation of concerns. Secured endpoints with JWT (access/refresh token rotation) with Redis-backed token blacklisting. Enforced role-based authorization across CLIENT, BARBER, and ADMIN workflows. Integrated Cloudinary for document uploads, AES-256-CBC encryption for sensitive fields (SSN, license numbers), and rate limiting with Redis caching on high-traffic endpoints. Designed admin approval pipeline for barber verification before platform access.
*Tech: NestJS, TypeScript, PostgreSQL, Prisma ORM, Redis, JWT, Cloudinary, Socket.IO, Stripe, Firebase*

**SmartFix – Home Maintenance Platform API.** Built a full-featured home maintenance platform backend using TypeScript and Express.js with MongoDB, supporting service requests, technician assignments, and booking management. Implemented JWT authentication with role-based access control across CLIENT, TECHNICIAN, and ADMIN roles. Integrated Firebase for push notifications, Stripe for payment processing, and Nodemailer for transactional emails. Added QR code generation for service verification and Cloudinary for media uploads. Deployed on Railway with a fully documented Postman collection. (GitHub)
*Tech: TypeScript, Node.js, Express.js, MongoDB, Mongoose, JWT, Socket.io, Firebase, Stripe, Cloudinary, Railway*

## EDUCATION

**Bachelor of Computer Science**, Tanta University — June 2026
Faculty of Computers and Information - Grade: Very Good | Graduation Project (SmartFix): Excellent
Relevant Coursework: Data Structures, Algorithms, Database Systems

**Backend Development Program**, Route Academy — 2025
Specialized training in Node.js, Express.js, NestJS, TypeScript, MongoDB, and REST APIs

## TECHNICAL TRAINING & ACHIEVEMENTS

- Completed Competitive Programming Training at Coach Academy – Advanced problem-solving and data structures
- Active contributor to open-source projects on GitHub with focus on backend technologies
- Participated in collaborative team projects emphasizing Agile methodologies and code review practices
- Building expertise in testing frameworks (Jest, Supertest) for API reliability and CI/CD pipelines

## LANGUAGES

**Arabic:** Native **English:** Professional Working Proficiency